



(b) The instruction set also includes these bit manipulation instructions:

Instruction		Explanation
Opcode	Operand	
AND	#n/Bn/&n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n/Bn/&n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n/Bn/&n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
LSL	#n	Bits in ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end.
<address> can be an absolute or a symbolic address # denotes a denary number, e.g. #123 B denotes a binary number, e.g. B01001010 & denotes a hexadecimal number, e.g. &4A		

The current content of the ACC is shown:

ACC	1	0	0	1	1	0	1	0
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- (i) The table has three sets of instructions. The binary number 10011010 is reloaded into the ACC before each set of instructions is run.

Complete the table by writing the content of the ACC after each set of instructions has run.

	Instructions	ACC content
1	LSL #2	
2	ADD #5 AND #30	
3	OR B11110010 INC ACC	

[3]

